



Distribution-Aware Adaptive Surrogate Gradients for Spiking Neural Networks

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Abstract. Spiking Neural Networks (SNNs) have gained significant attention for their biological plausibility and energy efficiency, yet their direct training via backpropagation is hindered by the non-differentiable nature of discrete spike generation. Existing surrogate gradient (SG) methods primarily rely on static shapes with fixed hyperparameters, which often fail to account for the non-stationary distribution of membrane potentials across different layers and training stages. This discrepancy leads to the critical “gradient mismatch” problem, causing training instability or sub-optimal convergence. To bridge this gap, we propose a practical Distribution-Aware Adaptive Surrogate Gradient (DA-ASG) method. By leveraging the real-time standard deviation of input features as a statistical guide, DA-ASG dynamically modulates the surrogate gradient width to better align with evolving potential distributions. We further incorporate a safety lock mechanism to constrain the modulation range, preventing numerical instability in extreme scenarios. Moreover, a gradient-based optimization strategy enables the network to automatically learn layer-specific gradient shapes. Extensive evaluations on static (CIFAR-10/100, Tiny-ImageNet) and neuromorphic (CIFAR-10-DVS) benchmarks demonstrate that DA-ASG effectively mitigates gradient vanishing and exploding issues. Experimental results demonstrate superior training stability across various architectures and time steps, achieving competitive or near-state-of-the-art performance in several settings.

Keywords: Spiking Neural Networks · Surrogate Gradient · Distribution-Aware Adaptive Surrogate Gradient

1 Introduction

Spiking Neural Networks (SNNs), often regarded as the third generation of neural networks, have attracted increasing attention in neuromorphic computing due to their biological plausibility and superior energy efficiency [1]. Unlike conventional Artificial Neural Networks (ANNs), which rely on continuous-valued activations, SNNs transmit information through discrete binary spikes, enabling event-driven computation and massive parallelism. However, the discrete spike-generation mechanism involves a non-differentiable Heaviside step function, whose derivative is zero almost everywhere and undefined at the firing threshold. This property blocks gradient propagation in standard backpropagation (BP), making direct training of deep SNNs challenging.

To address this issue, direct training methods based on Backpropagation Through Time (BPTT) have been widely adopted by unfolding SNNs along the temporal dimension [2]. Compared with early ANN-to-SNN conversion methods [6], which often suffer from high inference latency and limited compatibility with neuromorphic datasets, BPTT-based direct training enables end-to-end optimization of SNNs on complex tasks. Among these methods, Surrogate Gradient (SG) learning has become the dominant training paradigm [7]. By replacing the non-differentiable spike function with a smooth auxiliary derivative during the backward pass, SG allows SNNs to be optimized using gradient descent while preserving discrete spike-based computation in the forward pass.

Despite its effectiveness, most existing SG methods employ static surrogate functions with fixed shape parameters. Early studies explored various hand-designed surrogate derivatives, such as rectangular [3], exponential decay [7], triangular [8], sigmoid-based, and arctangent functions [9]. These methods usually rely on heuristically selected hyperparameters that remain unchanged across layers and training epochs. However, this “one-size-fits-all” strategy overlooks the intrinsic spatiotemporal dynamics of SNNs. In deep architectures such as ResNet, membrane potential distributions are highly non-stationary: they vary across different layers and continuously evolve as network weights are updated. A static surrogate function may therefore fail to align with these changing distributions, leading to the well-known gradient mismatch problem [10]. Specifically, an overly narrow surrogate gradient may cause gradient vanishing, whereas an overly broad one may introduce noisy or erroneous update signals. This mismatch can degrade training stability and convergence efficiency, especially in deep SNNs and dynamic neuromorphic scenarios.

Recent studies have attempted to improve SG learning by introducing dynamic or learnable gradient approximations. Annealing-based methods gradually adjust the surrogate shape during training. For example, Dspike [11] employs a temperature parameter to anneal the surrogate gradient toward a Dirac-delta-like function derived from finite-difference approximations, while gradual surrogate gradient learning [12, 13] progressively sharpens the gradient curve over training epochs. Learnable surrogate gradient methods further treat the surrogate width as a trainable parameter. A representative example is Learnable Surrogate Gradient (LSG) [4], which links the surrogate width to the learnable membrane decay factor and optimizes it through backpropagation. Search-based methods, such as Differentiable Hierarchical Search (DHS) [14], formulate surrogate function design as a differentiable search problem. Other approaches, such as Take a Shortcut Back (TSB) [15], improve gradient flow by modifying the backward path.

Although these methods improve upon static surrogate gradients, they still have several limitations. Annealing-based approaches often depend on manually designed schedules, while search-based methods introduce additional computational overhead. More importantly, many existing adaptive methods remain insensitive to real-time data distributions within individual training batches. Parameter-driven approaches such as LSG [4] typically evolve at the same timescale as synaptic weights, making them less responsive to instantaneous distribution shifts. In addition, they may rely on specific normalization techniques, such as Threshold-Dependent Batch Normalization (tdBN) [5], to enforce membrane potentials into predefined distributions, which restricts their flexibility across architectures and training conditions.

To bridge this gap, we propose a novel Distribution-Aware Adaptive Surrogate Gradient (DA-ASG) method that dynamically calibrates the surrogate gradient according to the statistical properties of neuronal activities. Instead of relying on fixed hyperparameters or slowly updated global parameters, DA-ASG leverages the online standard deviation of input features to modulate the width of the surrogate function in a layer-wise manner. This design establishes a data-driven feedback mechanism, enabling the surrogate gradient to better match the evolving range of membrane potentials during training. As a result, DA-ASG provides a more flexible and practical solution for alleviating gradient mismatch while reducing dependence on specific architectural or normalization priors.

The main contributions of this work are summarized as follows:

- We propose DA-ASG, a distribution-aware surrogate gradient learning method that addresses the mismatch between static surrogate gradients and dynamic membrane potential distributions in deep SNNs.
- We introduce a simple yet effective adaptive scaling mechanism that stabilizes the backward pass and alleviates gradient vanishing and exploding issues without requiring architectural redesign.
- Extensive experiments on static image classification benchmarks, including CIFAR-10, CIFAR-100, and Tiny-ImageNet, as well as the neuromorphic CIFAR-10-DVS dataset, demonstrate that DA-ASG achieves highly competitive performance. Compared with strong baselines, our method improves training stability and yields solid performance gains in several settings.

2 Preliminaries

SNNs process information over discrete time steps by transmitting binary spikes. In this work, we adopt the Leaky Integrate-and-Fire (LIF) neuron model, which is widely used in deep SNNs due to its computational efficiency and biological plausibility [16]. Given a simulation length T , the membrane potential of neuron i in layer l at time step t is updated as:

$$V_i^l[t] = V_i^l[t-1] + \frac{1}{\tau} \left(X_i^l[t] - (V_i^l[t-1] - V_{\text{reset}}) \right) \quad (1)$$

where τ is the membrane time constant, $X_i^l[t]$ denotes the presynaptic input current, and V_{reset} is the reset potential, which is typically set to 0.

When the membrane potential exceeds the firing threshold V_{th} , the neuron emits a binary spike $O_i^l[t]$:

$$O_i^l[t] = H(V_i^l[t] - V_{\text{th}}) = \begin{cases} 1, & \text{if } V_i^l[t] > V_{\text{th}} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

where $H(\cdot)$ denotes the Heaviside step function. After spike generation, a hard reset mechanism is applied to reset the membrane potential:

$$V_i^l[t] = V_i^l[t] \bullet (1 - O_i^l[t]) \quad (3)$$

The discontinuity of the Heaviside function makes the spike-generation process non-differentiable, preventing standard gradient-based optimization from being directly applied to SNNs. To address this issue, SG learning replaces the derivative of the step function with a smooth auxiliary function during backpropagation, while keeping the binary spike generation unchanged in the forward pass:

$$\frac{\partial O_i^l[t]}{\partial V_i^l[t]} \approx \sigma'(V_i^l[t] - V_{th}) \quad (4)$$

where $x = V_i^l[t] - V_{th}$ denotes the threshold-centered membrane potential, and $\sigma'(x)$ is the surrogate derivative. Most existing SG methods use fixed shape parameters, such as a constant scaling factor α , to control the gradient width. However, such static designs cannot adapt to the evolving membrane potential distributions across layers and training stages, which motivates our distribution-aware adaptive surrogate gradient method.

3 Method

In this section, we introduce the proposed DA-ASG method. As shown in Fig. 1, DA-ASG builds a statistical feedback loop between forward neuronal activity and backward gradient modulation. Specifically, the forward sensing layer estimates the real-time distribution of membrane potentials, the adaptive control layer maps the distribution statistics to a surrogate gradient shape parameter, and the backward modulation layer applies the adjusted surrogate gradient during backpropagation.

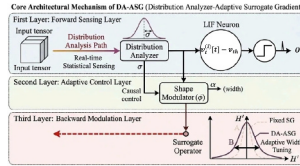


Fig. 1. The core architectural mechanism of the proposed DA-ASG. The framework is organized into three hierarchical layers: (1) Forward Sensing Layer captures the real-time statistical distribution (σ) of input features; (2) Adaptive Control Layer maps these statistics to the width control parameter α via the shape modulator ϕ ; (3) Backward Modulation Layer applies the customized surrogate gradient shape to the surrogate operator to achieve improved gradient alignment.

For the threshold-centered membrane potential $x = V - V_{th}$, we adopt the derivative of the sigmoid function as the surrogate gradient:

$$\sigma'(x) = \alpha \cdot \text{sigmoid}(\alpha x) \cdot (1 - \text{sigmoid}(\alpha x)) \quad (5)$$

where α controls the sharpness and effective support of the surrogate gradient. A larger α produces a sharper and narrower gradient curve, while a smaller α provides broader gradient coverage.

To adapt the surrogate gradient to the evolving membrane potential distribution, DA-ASG computes the raw adaptive scale as:

$$\alpha_{\text{raw}} = \frac{\alpha_{\text{base}}}{\sigma_{\text{std}}(x) \cdot \text{sigmoid}(k) + \epsilon} \quad (6)$$

where α_{base} is a layer-specific learnable parameter, $\sigma_{\text{std}}(x)$ denotes the instantaneous standard deviation of membrane potentials within a training batch, k is a learnable sensitivity factor, and ϵ is a small constant for numerical stability. In our implementation, α_{base} is initialized to 4.0, k is initialized to 1.0, and ϵ is set to 10^{-5} . This inverse modulation mechanism broadens the gradient coverage when the membrane-potential distribution is dispersed and sharpens the gradient when the distribution is concentrated. In implementation, $\sigma_{\text{std}}(x)$ is detached from the computational graph to avoid additional second-order gradient computation.

Although Eq. (6) enables data-dependent modulation, extremely small values of $\sigma_{\text{std}}(x)$ may lead to excessively large α_{raw} , causing unstable optimization. Therefore, we introduce a safety lock mechanism to constrain the adaptive scale within a valid range:

$$\alpha_{\text{da}} = \text{Clamp}(\alpha_{\text{raw}}, \alpha_{\text{min}}, \alpha_{\text{max}}) \quad (7)$$

we set $\alpha_{\text{min}} = 0.1$, $\alpha_{\text{max}} = 10.0$, which prevents the surrogate gradient from becoming either overly flat or excessively sharp under extreme input statistics.

Rather than treating α_{base} and k as fixed hyperparameters, DA-ASG formulates them as learnable parameters, denoted by $\Theta = \{\alpha_{\text{base}}, k\}$. These parameters are optimized jointly with synaptic weights through backpropagation:

$$\theta \leftarrow \theta - \eta \frac{\partial L}{\partial \theta}, \forall \theta \in \Theta \quad (8)$$

where L denotes the training loss, η is the learning rate, and θ represents each learnable parameter in Θ . This design allows each layer to learn its own gradient sensitivity while preserving responsiveness to batch-level distribution shifts.

Overall, DA-ASG provides a lightweight and data-driven mechanism for adaptive surrogate gradient modulation. By combining online distribution sensing, bounded shape control, and learnable layer-specific parameters, DA-ASG improves gradient stability without requiring architectural redesign and can be easily integrated into standard SNN training pipelines.

4 Experiments

To evaluate the effectiveness of DA-ASG, we conducted experiments on static image classification benchmarks, including CIFAR-10, CIFAR-100, and Tiny-ImageNet, as well as the neuromorphic CIFAR-10-DVS dataset. We first analyze gradient dynamics on CIFAR-10 to validate the training stability of DA-ASG, and then compare its classification accuracy with representative SNN training methods.

4.1 Gradient Dynamics and Stability Analysis

Experimental Setup for Gradient Analysis: Before reporting classification accuracy, we examine whether DA-ASG mitigates gradient mismatch and vanishing/exploding problems. We use a standard 6-layer Spiking CNN with four convolutional layers and two fully connected layers. The architecture is 64C3-128C3-P2-256C3-256C3-P2-512FC-10FC, where C3 denotes a 3×3 convolution and P2 denotes 2×2 average pooling. We track layer-wise gradient statistics over 200 training steps and compare DA-ASG with Sigmoid ($\alpha = 4.0$), Triangular (width = 1.0), Fast Sigmoid, and Arctan surrogate functions.

Global Training Stability: The global gradient norm reflects overall optimization stability. As shown in Fig. 2, static surrogate functions show different limitations. Arctan suffers from severe gradient explosion and is omitted from the plot for readability, while Triangular produces overly small gradients that may slow down weight updates. Although Sigmoid is relatively stable, it still exhibits gradient attenuation. In contrast, DA-ASG, shown by the blue curve, maintains the global gradient norm within a stable intermediate range of approximately 0.5 to 1.5. This indicates that the safety lock suppresses extreme gradient scaling, while distribution-aware modulation prevents gradient collapse.

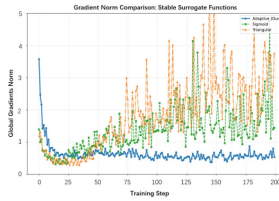


Fig. 2. Global Gradient Norm Comparison. DA-ASG (Blue) converges to a stable range, avoiding the decay observed in Sigmoid (Green) and the low magnitude of Triangular (Orange).

To further quantify stability, we count the number of layers with vanishing gradients ($< 10^{-6}$) and exploding gradients (> 1.0) at each training step. As shown in Fig. 3a, b static baselines frequently violate these stability bounds: Arctan causes explosions in shallow layers, whereas Sigmoid and Triangular tend to suffer from vanishing gradients in deeper layers. DA-ASG maintains almost zero exploding layers and only minimal vanishing cases throughout training, demonstrating more stable optimization behavior.

Layer-Wise Gradient Propagation: The “gradient vanishing” problem in SNNs typically manifests as signal decay in deeper layers. To investigate this, we analyzed the gradient distribution of weights at different depths: shallow (Conv1) and deep (Conv4).

We further analyze layer-wise gradient propagation by comparing gradients in shallow Conv1 and deep Conv4 layers. In Conv1, most stable methods preserve reasonable gradient ranges, and DA-ASG is comparable to Sigmoid. However, the difference becomes more pronounced in Conv4. As shown in Fig. 3c, the gradient ranges of Sigmoid and Triangular shrink to approximately ± 0.004 , indicating severe attenuation of

error signals. By contrast, DA-ASG, shown in blue, maintains a wider range of approximately ± 0.01 . This suggests that DA-ASG can adaptively increase gradient coverage in deeper layers, which is beneficial for training deeper SNN architectures.

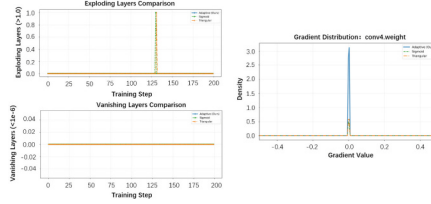


Fig. 3. Gradient stability and deep-layer gradient distribution analysis. (a) Vanishing layers. (b) Exploding layers. (c) Conv4 weight-gradient distribution. DA-ASG maintains stable layer-wise gradients and preserves a wider gradient distribution in the deeper layer compared with static surrogate functions.

4.2 Performance on Static Datasets

Table 1 summarizes the classification performance on CIFAR-10, CIFAR-100, and Tiny-ImageNet. The proposed DA-ASG method achieves competitive performance across various network architectures and simulation time steps.

Performance on CIFAR-10: On CIFAR-10, DA-ASG demonstrates competitive performance under low-latency settings and shows clearer advantages as the number of time steps increases. Under the ResNet-18 architecture, our method achieves 94.57%, 95.48%, and 95.55% accuracy at $T = 2, 4$, and 6, respectively. While ASGL [17] reports a higher accuracy at $T = 2$, DA-ASG achieves stronger performance at longer simulation horizons and remains competitive with recent neuron and training designs. Under ResNet-19, DA-ASG reaches 95.57% at $T = 6$, slightly surpassing LSG [4]. These results suggest that distribution-aware modulation is particularly effective when sufficient temporal integration is available, while maintaining competitive performance in low-latency regimes.

Performance on CIFAR-100: On the more challenging CIFAR-100 dataset, DA-ASG shows clearer advantages at moderate-to-longer simulation horizons. Using ResNet-18, our method reaches 78.69% accuracy at $T = 6$, exceeding CLIF [21] (78.36%) and ILIF [22] (78.51%), and also outperforming ASGL [17] under its reported $T = 4$ setting (77.74%). Similarly, on ResNet-19 at $T = 6$, DA-ASG reaches 78.25%, surpassing LSG [4] (76.32%) and TAB [19] (76.82%) under comparable settings. These results suggest that matching gradient width to the standard deviation of membrane potentials is particularly beneficial for more complex recognition tasks.

Results on Tiny-ImageNet: On the larger Tiny-ImageNet dataset, DA-ASG further demonstrates promising generalization capability. Using the VGG-13 architecture, our method achieves 57.44% accuracy at $T = 8$. This exceeds ASGL [17] (56.81% at the same T) and is also higher than Offline LTL [24] (55.37%), despite the latter using a

Table 1. Comparison of classification performance on static datasets.

Dataset	Method	Architecture	Accuracy (%) Timestep (2/4/6)		
CIFAR-10	LSG [4]	ResNet-19	94.41	95.17	95.52
	ASGL [17]	ResNet-18	95.27	95.35	N/A
	SLTT [18]	ResNet-18	N/A	N/A	94.59
	ShortcutBP [15]	ResNet-18	93.89	94.30	N/A
	TAB [19]	ResNet-19	94.73	94.76	94.81
	PSG [20]	ResNet-19	N/A	95.12	95.00
	CLIF [21]	ResNet-18	N/A	94.89	95.41
	ILIF [22]	ResNet-18	N/A	95.24	95.49
	Our Method	ResNet-19	94.27	95.49	95.57
	Our Method	ResNet-18	94.57	95.48	95.55
CIFAR-100	LSG [4]	ResNet-19	76.32	76.85	76.32
	ASGL [17]	ResNet-18	76.59	77.74	N/A
	SLTT [18]	ResNet-18	N/A	N/A	74.67
	ShortcutBP [15]	ResNet-18	73.68	74.78	N/A
	TAB [19]	ResNet-19	76.31	76.81	76.82
	PSG [20]	ResNet-19	N/A	75.72	N/A
	IM-LIF [23]	ResNet-19	N/A	77.21	77.42
	CLIF [21]	ResNet-18	N/A	77.00	78.36
	ILIF [22]	ResNet-18	N/A	77.43	78.51
	Our Method	ResNet-19	76.16	77.83	78.25
	Our Method	ResNet-18	76.18	77.93	78.69
Tiny-ImageNet	ASGL [17]	VGG-13	56.57(T = 4)	56.81(T = 8)	N/A
	Offline LTL [24]	VGG-13	N/A	N/A	55.37(T = 16)
	IM-LIF [23]	ResNet-19	54.82(T = 3)	55.37(T = 6)	N/A
	Our Method	VGG-13	56.48(T = 4)	57.44(T = 8)	N/A

much larger time step ($T = 16$). These results indicate that the safety lock mechanism and adaptive scaling remain effective on deeper networks with high-dimensional inputs.

The relatively modest improvement on CIFAR-10 may be attributed to its lower task complexity. Since CIFAR-10 contains fewer categories and exhibits relatively stable feature distributions, conventional static surrogate gradients can already capture the main neuronal dynamics. In contrast, more challenging datasets such as CIFAR-100 and Tiny-ImageNet amplify the non-stationarity of membrane potentials, making static surrogate gradients more susceptible to gradient mismatch. DA-ASG therefore shows clearer benefits in deeper architectures and more complex tasks by preserving effective gradient flow and improving representation learning.

4.3 Performance on Neuromorphic Datasets

We evaluated DA-ASG on the event-driven CIFAR-10-DVS dataset using the VGG-11 architecture to test its adaptability to sparse, asynchronous data. As shown in Table 2, DA-ASG achieves an accuracy of 78.9% at $T = 10$. This result outperforms recent efficient training methods such as SLTT [18] (77.3%) and NDOT [25] (77.5%), as well as the advanced ILIF model [22] (78.6%). The result on neuromorphic data indicates that utilizing real-time statistical properties is effective for handling the non-stationary dynamics inherent in event streams.

Table 2. Performance comparison CIFAR-10-DVS

Method	Architecture	Timestep	Accuracy (%)
OTTT [26]	VGG-11	10	76.3
SLTT [18]	VGG-11	10	77.3
ILIF [22]	VGG-11	10	78.6
NDOT [25]	VGG-11	10	77.5
Our Method	VGG-11	10	78.9

4.4 Ablation Study

To assess the contribution of each component in DA-ASG, we conduct ablation studies on CIFAR-10 and CIFAR-100. ResNet-19 is used for CIFAR-10, while ResNet-18 is used for CIFAR-100 to cover different architectural depths and task complexities. The results are reported in Table 3.

Impact of Adaptive Surrogate Gradients: We first compare DA-ASG with standard static surrogate functions. As shown in the first two rows of Table 3, Rectangular and Static Sigmoid achieve baseline accuracies of about 93.95–94.62% on CIFAR-10 and 76.23% on CIFAR-100. Introducing a learnable α , which is optimized by backpropagation but independent of input statistics, improves performance by 0.72% on CIFAR-10 and 1.06% on CIFAR-100 compared with Static Sigmoid. This indicates that fixed surrogate shapes are suboptimal for deep SNNs, and that adaptive gradient scaling is beneficial for optimization.

Effectiveness of Distribution-Awareness: We then evaluate the effect of distribution-aware modulation, which is the core component of DA-ASG. Compared with Learnable α without statistics, the full DA-ASG further improves accuracy to 95.49% on CIFAR-10 and 77.93% on CIFAR-100. This demonstrates that adapting the surrogate shape according to the real-time standard deviation of membrane potentials can better alleviate gradient mismatch than using learnable parameters alone.

Synergy with Normalization Techniques: Finally, we examine the compatibility of DA-ASG with different normalization strategies. DA-ASG with standard Batch Normalization already achieves competitive results, reaching 95.44% on CIFAR-10 and 77.71% on CIFAR-100. Replacing BN with tdBN [5] further improves the performance to 95.49% and 77.93%, respectively. This suggests that tdBN provides more stable temporal membrane potential statistics, allowing DA-ASG to estimate distribution-aware gradient scales more effectively. Since tdBN is widely used in directly trained SNNs, this compatibility highlights the practicality of DA-ASG in mainstream SNN training pipelines.

Table 3. Ablation study on component effectiveness

Method	Learnable Shape	Dist. Aware	Norm	CIFAR-10 (ResNet-19)	CIFAR-100 (ResNet-18)
Rect Function	×	×	tdBN	93.95%	76.23%
Static Sigmoid	×	×	tdBN	94.62%	76.23%
Learnable α	✓	×	tdBN	95.34%	77.29%
Ours (DA-ASG)	✓	✓	BN	95.44%	77.71%
Ours (DA-ASG)	✓	✓	tdBN	95.49%	77.93%

5 Conclusion

This paper presents the Distribution-Aware Adaptive Surrogate Gradient (DA-ASG), a practical optimization strategy designed to mitigate the gradient mismatch problem in deep Spiking Neural Networks (SNNs). By dynamically modulating the surrogate gradient width based on real-time membrane potential statistics and incorporating a safety lock mechanism, DA-ASG better aligns backward gradient flow with the evolving spatiotemporal dynamics of neuronal activities. Extensive evaluations on both static (CIFAR-10/100, Tiny-ImageNet) and neuromorphic (CIFAR-10-DVS) benchmarks show that DA-ASG improves training stability and delivers competitive, and in several settings superior, classification performance. Rather than claiming universal gains across all configurations, our results indicate that distribution-aware surrogate adaptation is especially beneficial for deeper SNNs, more challenging datasets, and moderate-to-longer simulation horizons. Future research will explore higher-order statistical descriptors for more granular modulation and the hardware-efficient implementation of this adaptive framework on low-power neuromorphic platforms.

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